

COM S 127 - Lab Week 6 Grading Rubric

This lab was assigned for week 6 of the Fall semester.

This lab is due by 11:59 p.m. of the sixth day after the student's initial lab meeting day.

Lab Objective

The purpose of this lab is for students to review the material presented in Exam #1, and to begin experimenting with 'while loops.'

Instructions

Submission 'Check Offs'

For students to receive credit for their lab work, they must 'check off' their exercises with either the Graduate TA or Undergraduate TA (UGTA) for their lab section. If the lab cannot be completed during the assigned lab hours, the student **MUST** either attend a **Graduate TAs** office hours to complete the 'check off' process, or arrange for a Zoom/ WebEx meeting with a Graduate TA to receive credit for their work.

This means that **STUDENTS WILL NO LONGER BE ABLE TO SUBMIT THEIR WORK ON CANVAS**. As such, **all work MUST 'checked off' in person or via a Zoom/ WebEx appointment**

There is a list of Graduate TA office hours and email addresses at the top of the syllabus on Canvas.

Important Notes:

All scripts should include the student's name and the date they programmed the script on the top line of the file. The student should also include the week of the lab, and the exercise number on the second line. Example:

```
# <Student Name>           <The Date>
# Lab Week <week of the lab> - Exercise #<exercise number>

# Matthew Holman           9-26-2022
# Lab Week 6 - Exercise #2
```

Readings:

keywords (advanced)

- 2.5. Variable Names and Keywords

while loops

- 8.3. The while Statement

- 8.8. Other uses of while

Lab Challenge Activities:

Create a single script called **examReview.py**, which **prints** out the results of the Python statements found in the **labWeek6WhileLoopsExamReview.pptx** file.

Please feel free to copy/ paste the code of these questions into VS Code. Your hand-computed answers should replace the "?" strings in the .format method calls in each print statement.

- To facilitate learning, students should attempt to determine the output by hand before programming/ running the script, and should print out their initial hand computed answer.
 - **THIS INCLUDES EVALUATING EACH EXPRESSION AND PRINTING WHAT EACH OF THESE EXPRESSIONS EVALUATE TO.**
 - Please note: the student's hand computed output is just for learning purposes, and will not be evaluated as a part of the grade. Meaning - if the student gets an incorrect hand computed result, that will not impact the final grade.
- Students should also print out the answer the computer provides.
 - See the **Example Output** section for details.
- Please note: **all** of the exercises from **this** portion of the lab should appear in the **examReview.py** file. The exercises **below** should each appear **in their own files**.

Create a script called **exercise1.py**, such that it completes the task in **Exercise 1** on **slide 7** of the **labWeek6WhileLoopsExamReview.pptx** file.

Create a script called **exercise2.py**, such that it completes the task in **Exercise 2** on **slide 9** of the **labWeek6WhileLoopsExamReview.pptx** file.

Deliverables:

Show the table of contents of the Runestone textbook to a Graduate TA/ UGTA with green 'check marks' next to the relevant sections noted in the **Readings** section of this document once that reading is complete.

Create a script called **examReview.py** which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called **exercise1.py** which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

Create a script called **exercise2.py** which completes the tasks laid out in the **Lab Challenge Activities** section of this document.

All the deliverable scripts and the Runestone table of contents should be shown to a Graduate TA/ UGTA

Resources:

Official Python Tutorial (First Steps Towards Programming - introduces while loops):
<https://docs.python.org/3/tutorial/introduction.html#first-steps-towards-programming>

Official Python Tutorial (break and continue Statements):

<https://docs.python.org/3/tutorial/controlflow.html#break-and-continue-statements-and-else-clauses-on-loops>

Official Python Tutorial (pass Statements): <https://docs.python.org/3/tutorial/controlflow.html#pass-statements>

Files Provided

- labWeek6WhileLoopsExamReview.pptx

Example Code/ Output

examReview.py output:

```
Question 1
c = a % b | H: 7 C: 2
c *= 2 | H: 2 C: 4
```

etc.

exercise1.py output:

```
Enter an integer: 10
10
8
6
4
2
```

exercise2.py output:

```
Enter an integer: 3
Enter an integer: 4
12 9 6 3
8 6 4 2
4 3 2 1
```

Grading Items

- **(Attendance)** Was the student present at the lab or had made arrangements to attend virtually?: _____ / 5
- **(Runestone Academy Reading)** Has the student demonstrated that they have completed their Runestone Academy reading by showing a Graduate TA/ UGTA the table of contents with green 'check marks' next to the relevant completed sections?: _____ / 5
- **(examReview.py)** Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section and the **Example Code/ Output** section?: _____ / 10
- **(exercise1.py)** Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section?: _____ / 5

- **(exercise2.py)** Did the student successfully complete a script which generates correct output as per the **Lab Challenge Activities** section?: _____ / 5

TOTAL _____ / 30